Case Study on the Analysis of SDLC Models Applied to a Real-World Software Application

AIM

To analyze the Software Development Life Cycle (SDLC) models used in the development of a real-world software application and identify the most suitable SDLC model.

OBJECTIVE

- Study different SDLC models.
- Analyze the development process of a real-world software application.
- Compare the advantages and limitations of various SDLC models.
- Recommend the most appropriate SDLC model for the application.

SELECTED SOFTWARE APPLICATION

Supply Chain & Logistics: Vendor Onboarding & Contract Verification Hub

OVERVIEW

A Vendor Onboarding & Contract Verification Hub allows enterprise organizations to:

- Register and complete secure profile onboarding.
- Upload mandatory legal compliance, corporate tax, and operational certification documents.
- Route contracts through parallel corporate department approval gates.
- Conduct digital signature executions and securely process background validation audits.
- Automatically track compliance renewals, SLAs, and performance metrics.
- Provide data reports, analytics, and active risk scoring dashboards.

FUNCTIONAL REQUIREMENTS

Vendor Ingestion Module

- Vendor Registration and Authentication Portal
- Secure Document Attachment Box (Tax Certificates, Insurance Proofs)
- Corporate Profile & Operational Capacity Input Forms

Verification & Document Processing Layer

- Parallel Workflow Router (Legal, Compliance, and Procurement)
- Digital Document Signature Execution (E-sign Integration)
- Compliance Validity Verification & Renewal Watchers

Risk & Procurement Module

- Automated Background Audit Check Trigger
- Tiered Discount & Contract Terms Override Logic
- Vendor Performance Rating & Service Level Agreement Tracking

Administrative Hub

- System Security and Access Authority Control
- Corporate Procurement Spend & Capacity Analytics Sheets
- Multi-Role Trend Summary Reports

SDLC MODELS ANALYSIS

SDLC Model	Suitability	Advantages	Limitations
Waterfall	Low	Simple structure; easy to map baseline requirements.	Extremely rigid; struggles to modify evolving compliance requirements.
Prototype	Medium	Enables clear client validation of early dashboard designs.	Increases initial capital and development hours significantly.
Spiral	High	Provides rigorous risk assessments and compliance protection.	Highly complex structure; excessive implementation timeline.
Agile	Very High	Supports iterative configuration loops and immediate enhancements.	Demands constant business alignment and high operational expertise.
DevOps	Very High	Ensures robust automated testing and continuous pipeline deployment.	Demands complex continuous orchestration configurations.

WHY AGILE IS BEST?

Reasons:

- Compliance mandates and operational validation requirements modify frequently.
- New parallel approval gates and custom corporate modules must be added regularly.
- Document verification bug patches and logic extensions must update instantly.
- Enterprise workflows require frequent platform and cloud connectivity revisions.
- Active administrative feedback directly guides ongoing interface usability updates.

AGILE DEVELOPMENT PROCESS

Requirement Gathering → Sprint Planning → Design → Development → Testing → Deployment → Customer Feedback → Next Sprint

SDLC PHASES

1. Requirement Analysis

Collect requirements for profile ingestion forms, parallel legal gates, secure e-signature APIs, and OCR documentation scanners. *Deliverables:* Software Requirement Specification (SRS) document.

2. System Design

Design multi-role structural access layouts, document mapping database logic, workflow blueprints, and webhook data maps. *Deliverables:* Database Schemas and UML Interaction Models.

3. Development

Construct profile registration panels, compliance rules systems, parallel signature tracking modules, and notification routers. Developed via multi-role programming framework loops bound to secure relational tables.

4. Testing

Conduct automated system integration runs, parallel logic loop triggers, cross-department role validation, and User Acceptance Testing (UAT).

Test Case	Expected Result
Vendor Form Profile Creation	Successful Profile Generation and Database Log Creation
Omission of Required Compliance File	Validation Interception Block and Missing File Error Alert
Parallel Approval Route Initiation	Simultaneous Multi-Department Workspace Notification Delivery
Digital E-Sign Approval Triggers	Instant Contract Ingestion Lock and Verification State Status Shift

5. Deployment

Execute cloud deployment configurations, multi-portal application updates, and secure cloud platform server initializations.

6. Maintenance

Perform urgent logic revisions, compliance parameter revisions, workflow performance balancing, and API connection maintenance.

COMPARISON OF SDLC MODELS

Criteria	Waterfall	Prototype	Spiral	Agile	DevOps
Req. Changes	Poor	Good	Excellent	Excellent	Excellent
Feedback Cycles	Low	High	High	Very High	Very High
Risk Mitigation	Low	Medium	High	Medium	High
Deployment Speed	Slow	Medium	Medium	Fast	Very Fast
Capital Cost	Low	Medium	High	Medium	Medium
Hub Suitability	No	Partially	Yes	Best	Best

RESULTS

The platform dictates continuous operational revisions and immediate legal feedback. Agile development loop frameworks guarantee iterative setup velocity, deployment safety, and flexible workflows. Concurrently, DevOps continuous automation practices complement the structure by providing automated verification, document security updates, and instant code testing configurations. Therefore, an integrated Agile with DevOps paradigm serves as the optimal framework for deploying a secure, resilient vendor onboarding portal.

CONCLUSION

This case study clearly demonstrates that choosing the proper engineering framework depends entirely on specific application metrics. For compliance-critical, highly dynamic software matrices such as vendor registration and contract verification hubs, Agile processes secure system flexibility, swift deployment cycles, and robust stakeholder interaction, while DevOps automation principles maximize cloud deployment safety and data pipeline efficiencies. This powerful methodology intersection secures high enterprise software quality, rapid structural adaptations, and enhanced corporate team satisfaction.